Solution

Matrix Theory, Quiz 7C — April 10, 2026

For the matrix $A = \begin{bmatrix} 3 & 6 \\ -1 & -4 \end{bmatrix}$, answer the following.

3 (1) Find the characteristic polynomial of A.
1 (2) Find the eigenvalues of A.
4 (3) For each eigenvalue, find the eigenvector.
2 (4) True or False: A is diagonalizable.
If true, find the matrix P and the diagonal matrix D such that $A = PDP^{-1}$.

(1) $(3-\lambda)(-4-\lambda)+6 = \lambda^2+\lambda-6 = (\lambda+3)(\lambda-2)$

(2) $(\lambda_1, \lambda_2) = (-3, 2)$

(3) $\lambda_1 = -3 \quad A+3 = \begin{bmatrix} 6 & 6 \\ -1 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad v_1 = c\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$\lambda_2 = 2 \quad A-2 = \begin{bmatrix} 1 & 6 \\ -1 & -6 \end{bmatrix} \to \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} \quad v_2 = c\begin{pmatrix} -6 \\ 1 \end{pmatrix}$

(4) True. $P = [v_1 \ v_2] = \begin{bmatrix} 1 & -6 \\ -1 & 1 \end{bmatrix} \quad D = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$

<u>Version 7D</u>: (1) $A = \begin{pmatrix} -4 & -2 \\ 7 & 5 \end{pmatrix}$ char poly $= (-4-\lambda)(5-\lambda)+14 =$

$\lambda^2-\lambda-6 = (\lambda-3)(\lambda+2)$

(2) $(\lambda_1, \lambda_2) = (3, -2)$

(3) $\lambda_1 = 3 \quad A-3 = \begin{bmatrix} -7 & -2 \\ 7 & 2 \end{bmatrix} \to \begin{bmatrix} 7 & 2 \\ 0 & 0 \end{bmatrix} \quad v_1 = c\begin{pmatrix} -2 \\ 7 \end{pmatrix}$

$\lambda_2 = -2 \quad A+2 = \begin{bmatrix} -2 & -2 \\ 7 & 7 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \quad v_2 = c\begin{pmatrix} 1 \\ -1 \end{pmatrix}$

(4) True $P = \begin{bmatrix} -2 & 1 \\ 7 & -1 \end{bmatrix} \quad D = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$